deserts_hs_analyses_2

Evan Yi

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Set Up

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
desert_threshold <- 0.33

# Loading csv files
poverty_data <- read_csv("poverty_total_and_under5_clean.csv")
```

```
## Rows: 84338 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): GEOID, NAME, flag
## dbl (2): pct_poverty_total, pct_poverty_under5
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
childcare_data <- read_csv("cap_ccaccess_2025_revised_02282026.csv")
```

```
## Rows: 2281139 Columns: 14
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): state_fips, state_name, state_abbrv, county_fips, county_name, scho...
## dbl (7): longitude, latitude, adj_supply, adj_supply_fcc, adj_supply_ccc, ad...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
joined_rural <- read_csv("joined_rural.csv")
```

```
## Rows: 2281139 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (9): state_name, state_abbrv, county_fips, county_name, schooldistrict_c...
## dbl (9): longitude, latitude, adj_supply, adj_supply_fcc, adj_supply_ccc, ad...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data cleaning

```r
# Cleaning the poverty data
poverty_county <- poverty_data %>%
  # Extract 5-digit county FIPS from the 11-digit GEOID
  mutate(county_fips = str_sub(GEOID, 1, 5)) %>%
  filter(!is.na(pct_poverty_under5)) %>%
  group_by(county_fips) %>%
  summarise(
    avg_pct_poverty_under5 = mean(pct_poverty_under5, na.rm = TRUE),
    .groups = "drop"
  )

# Cleaning and joining the childcare data
childcare_with_poverty <- childcare_data %>%
  # Ensure county_fips is a 5-character string with leading zeros
  mutate(county_fips = str_pad(as.character(county_fips), width = 5, pad = "0")) %>%
  left_join(poverty_county, by = "county_fips")

# Cleaning rurality data
rural_lookup <- joined_rural %>%
  select(longitude, latitude, rural) %>%
  distinct()

# Joining rurality data
childcare_with_poverty <- childcare_with_poverty %>%
  left_join(rural_lookup, by = c("longitude", "latitude"))

# Check join quality
cat("Rows with rural status:", sum(!is.na(childcare_with_poverty$rural)), "\n")
```

```
## Rows with rural status: 2277112
```

```r
cat("Rows missing rural status:", sum(is.na(childcare_with_poverty$rural)), "\n")
```

```
## Rows missing rural status: 4027
```

```r
# Drop missing values and code new variables
childcare_with_poverty <- childcare_with_poverty %>%
  filter(!is.na(adj_supply_hs), !is.na(avg_pct_poverty_under5), !is.na(rural)) %>%
  mutate(
    # Estimated eligible (poverty) children per row out of 10 synthetic children
    eligible_children = 10 * (avg_pct_poverty_under5 / 100),
    # Desert indicator for Head Start
    hs_desert = adj_supply_hs <= desert_threshold
  )
```

Calculating eligibility

VERSION NOTE: As expected, two values were adjusted slightly due to universal filtering at the top of the analysis.

total_eligible_children: 3854947 -> 3834565 desert_eligible_children: 3820193 -> 3799876 pct_eligible_in_desert: 99.1 -> 99.1

```r
hs_eligible_desert <- childcare_with_poverty %>%
  summarise(
    total_eligible_children  = sum(eligible_children),
    desert_eligible_children = sum(eligible_children[hs_desert]),
    pct_eligible_in_desert   = round((desert_eligible_children / total_eligible_children) * 100, 1)
  )

print(hs_eligible_desert)
```

```
## # A tibble: 1 x 3
##   total_eligible_children desert_eligible_children pct_eligible_in_desert
##                     <dbl>                    <dbl>                  <dbl>
## 1                3834565.                 3799876.                   99.1
```

Calculating eligibility by rurality

The values seemed very high, so a follow-up analysis was conducted that looked at rurality.

VERSION NOTE: As expected, no values changed.

```r
hs_eligible_desert_by_rural <- childcare_with_poverty %>%
  group_by(rural) %>%
  summarise(
    total_eligible_children  = sum(eligible_children),
    desert_eligible_children = sum(eligible_children[hs_desert]),
    pct_eligible_in_desert = round((desert_eligible_children / total_eligible_children) * 100, 1),
  )

print(hs_eligible_desert_by_rural)
```

```
## # A tibble: 2 x 4
##   rural total_eligible_children desert_eligible_children pct_eligible_in_desert
##   <dbl>                   <dbl>                    <dbl>                  <dbl>
## 1     0                3099467.                 3089368.                   99.7
## 2     1                 735098.                  710509.                   96.7
```

Calculating eligibility by rurality and degree of scarcity

The values still seemed very high even with the rurality analysis. A follow-up analysis was conducted that looked at the degree of scarcity and confirmed parameters.

VERSION NOTE: As expected, no values changed.

```
# Weighted mean and median supply by rurality
childcare_with_poverty %>%
  group_by(rural) %>%
  summarise(
    weighted_mean_supply = round(
      weighted.mean(adj_supply_hs, w = eligible_children, na.rm = TRUE), 4
    ),
    median_supply = round(median(adj_supply_hs), 4),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 3
##   rural weighted_mean_supply median_supply
##   <dbl>                <dbl>         <dbl>
## 1     0               0.0254         0.003
## 2     1               0.0564         0
```

```
# Breaking down the share of areas with some supply by rurality
childcare_with_poverty %>%
  mutate(
    supply_category = case_when(
      adj_supply_hs == 0                ~ "No supply",
      adj_supply_hs <= desert_threshold ~ "Some supply, still a desert",
      TRUE                              ~ "Not a desert"
    )
  ) %>%
  group_by(rural, supply_category) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(rural) %>%
  mutate(pct = round(n / sum(n) * 100, 1)) %>%
  arrange(rural, supply_category)
```

```
## # A tibble: 6 x 4
## # Groups:   rural [2]
##   rural supply_category                   n   pct
##   <dbl> <chr>                         <int> <dbl>
## 1     0 No supply                    397294  20.7
## 2     0 Not a desert                   4665   0.2
## 3     0 Some supply, still a desert 1521182  79.1
## 4     1 No supply                    171921  48.6
## 5     1 Not a desert                   9547   2.7
## 6     1 Some supply, still a desert  172503  48.7
```